

Mobile IP

Motivation for Mobile IP

❑ Routing

- o based on IP destination address, network prefix (e.g. 129.13.42) determines physical subnet
- o change of physical subnet implies change of IP address to have a topological correct address (standard IP) or needs special entries in the routing tables

❑ Specific routes to end-systems?

- o change of all routing table entries to forward packets to the right destination
- o does not scale with the number of mobile hosts and frequent changes in the location, security problems

❑ Changing the IP-address?

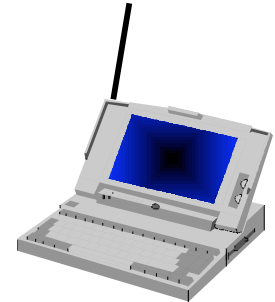
- o adjust the host IP address depending on the current location
- o almost impossible to find a mobile system, DNS updates take too much time
- o TCP connections break, security problems

Mobile IP Requirements

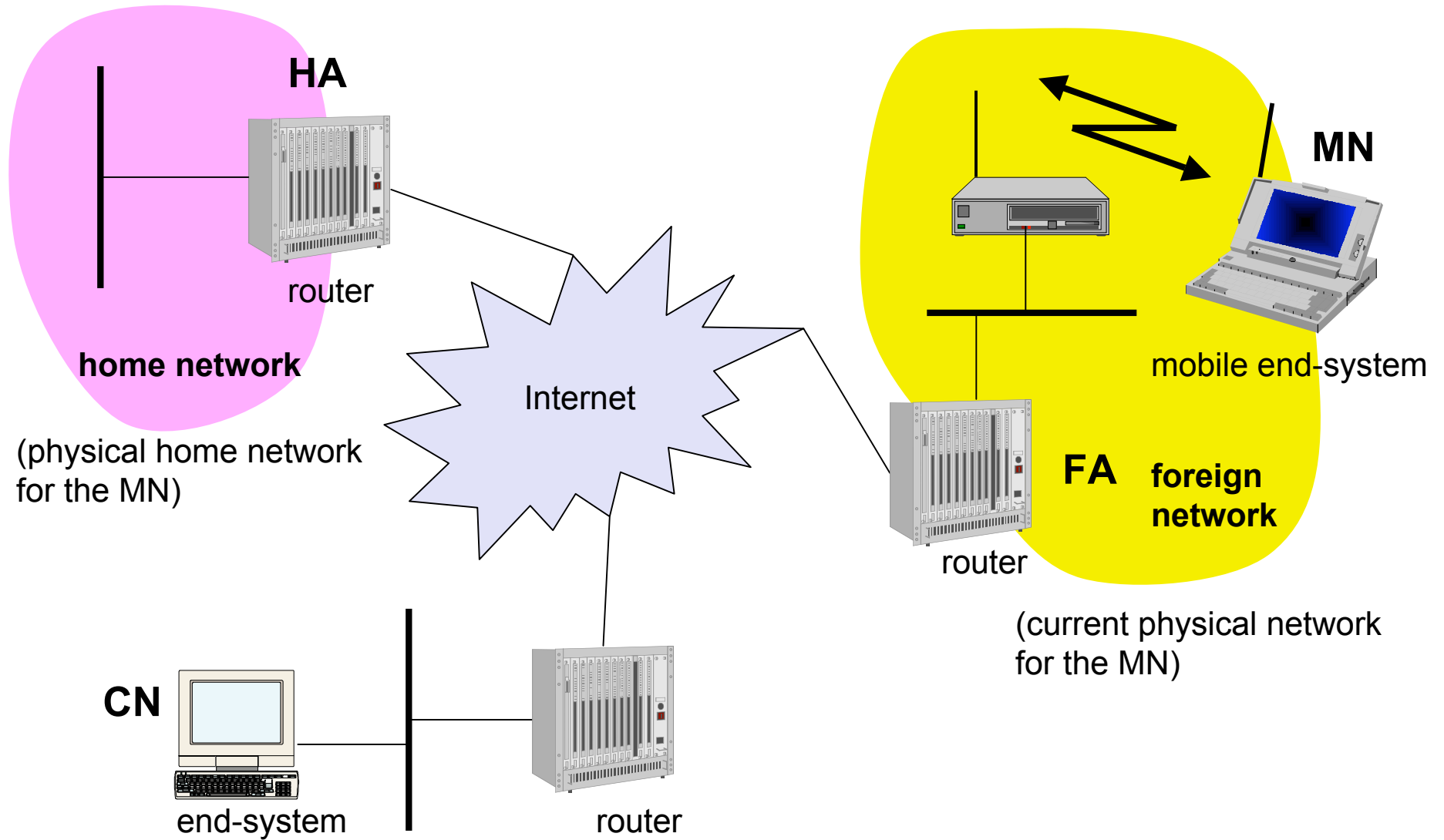
- ❑ Transparency
 - o mobile end-systems keep their IP address
 - o continuation of communication after interruption of link possible
 - o point of connection to the fixed network can be changed
- ❑ Compatibility
 - o support of the same layer 2 protocols as IP
 - o no changes to current end-systems and routers required
 - o mobile end-systems can communicate with fixed systems
- ❑ Security
 - o authentication of all registration messages
- ❑ Efficiency and scalability
 - o only little additional messages to the mobile system required (connection typically via a low bandwidth radio link)
 - o world-wide support of a large number of mobile systems in the whole Internet

Terminology

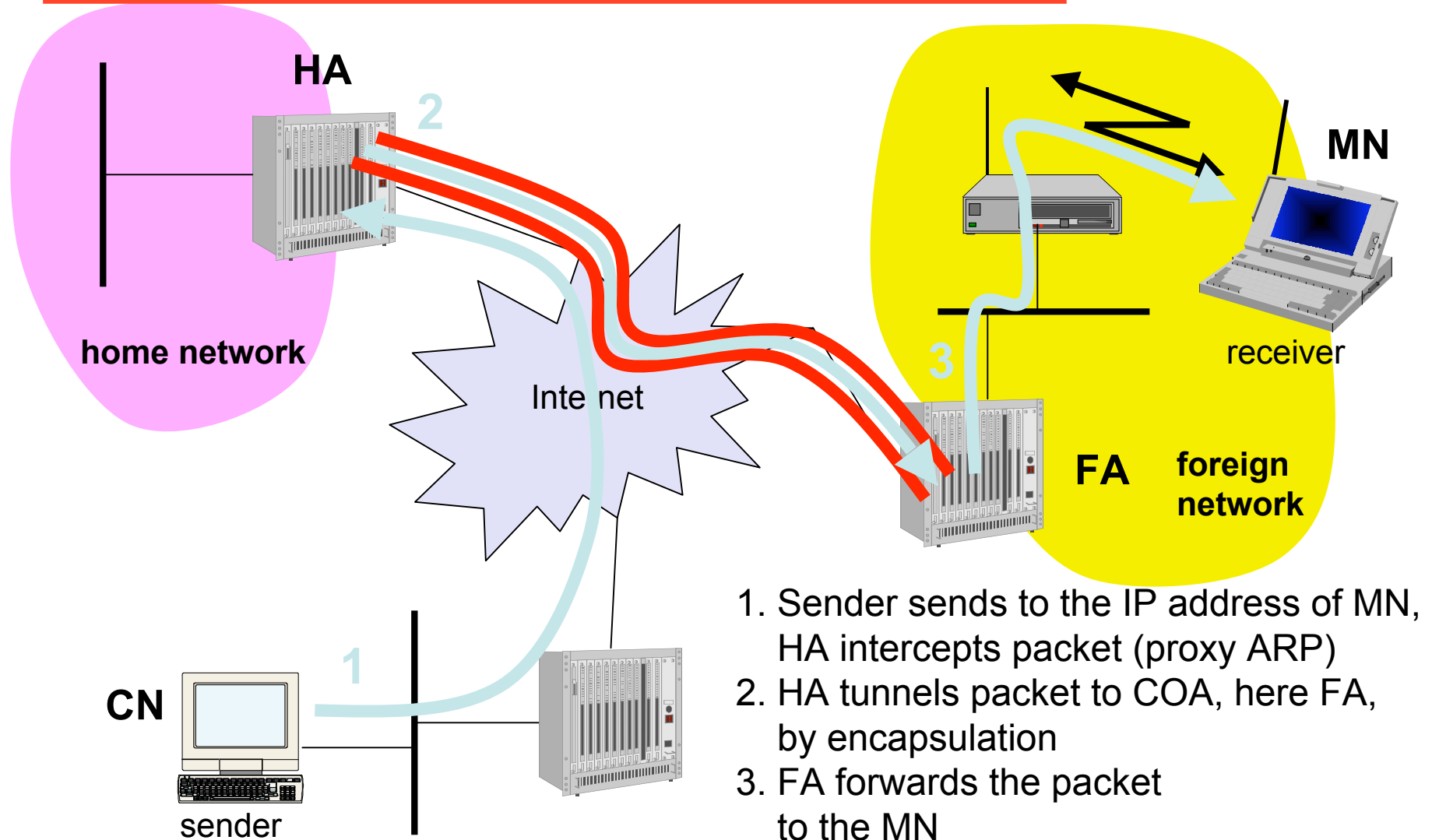
- ❑ Mobile Node (MN)
 - o system (node) that can change the point of connection to the network without changing its IP address
- ❑ Home Agent (HA)
 - o system in the home network of the MN, typically a router
 - o registers the location of the MN, tunnels IP datagrams to the COA
- ❑ Foreign Agent (FA)
 - o system in the current foreign network of the MN, typically a router
 - o forwards the tunneled datagrams to the MN, typically also the default router for the MN
- ❑ Care-of Address (COA)
 - o address of the current tunnel end-point for the MN (at FA or MN)
 - o actual location of the MN from an IP point of view
 - o can be chosen, e.g., via DHCP
- ❑ Correspondent Node (CN)
 - o communication partner



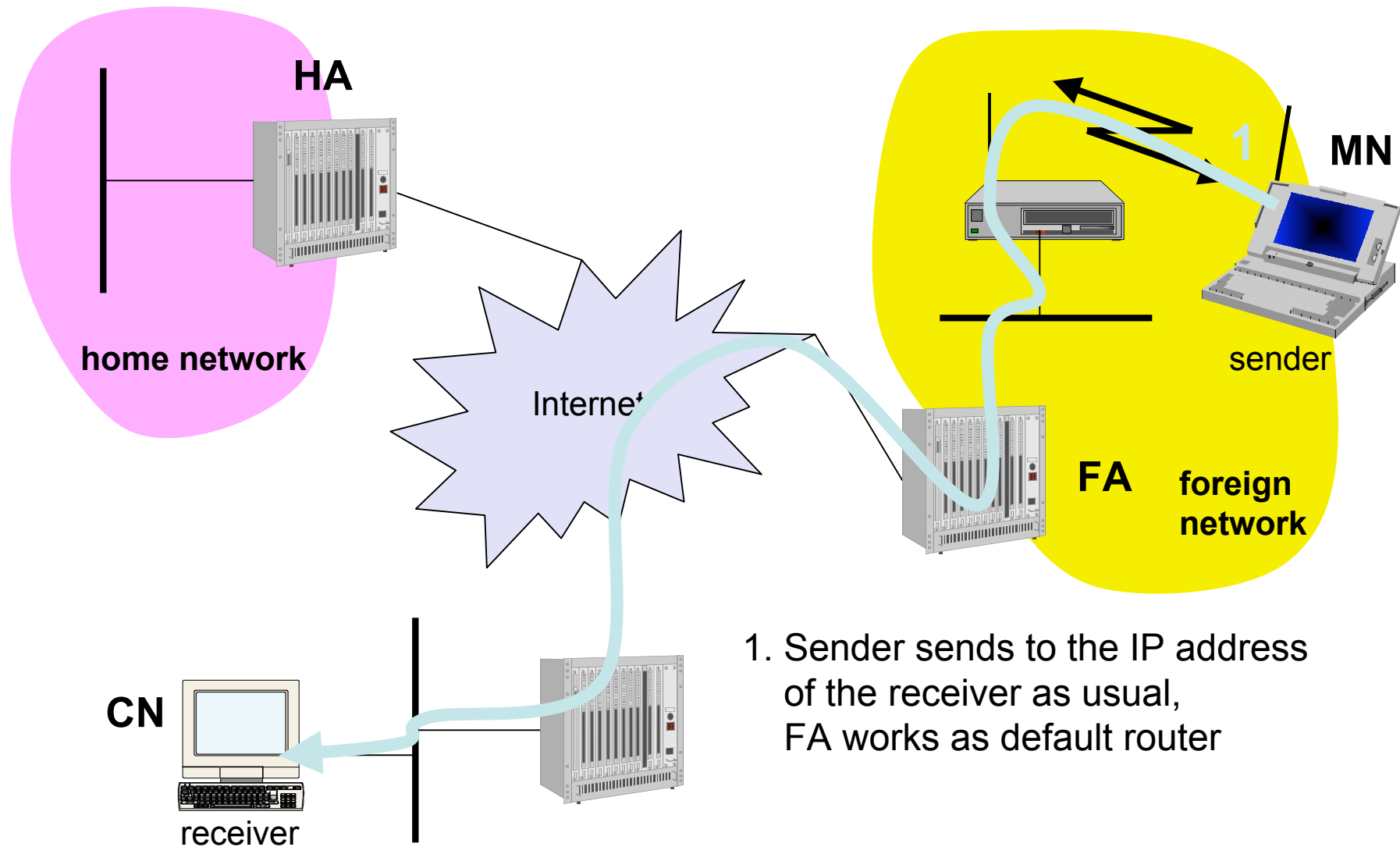
Example network



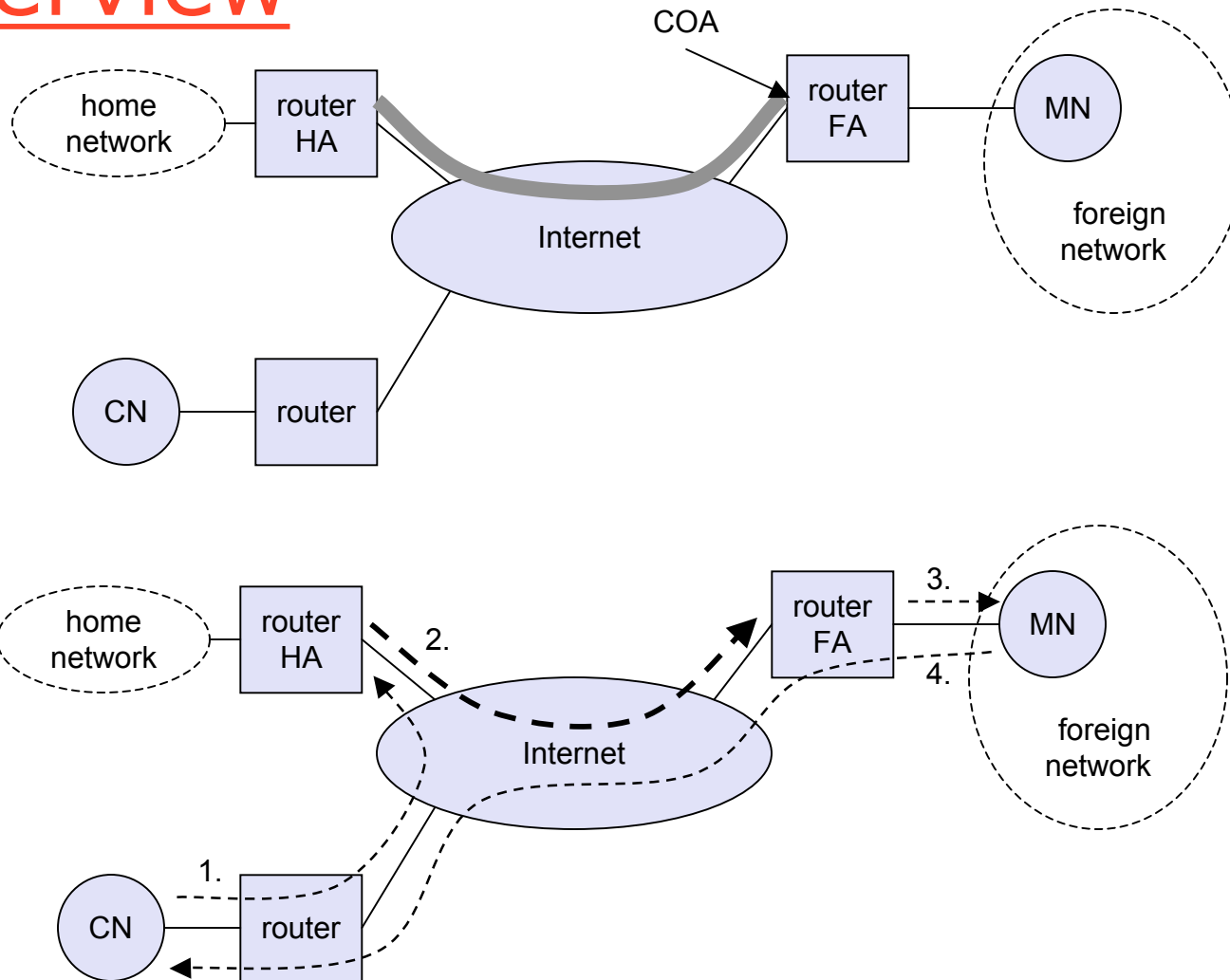
Data transfer to the mobile



Data transfer from the mobile



Overview



Network integration

❑ Agent Advertisement

- o HA and FA periodically send advertisement messages into their physical subnets
- o MN listens to these messages and detects, if it is in the home or a foreign network (standard case for home network)
- o MN reads a COA from the FA advertisement messages

❑ Registration (always limited lifetime!)

- o MN signals COA to the HA via the FA, HA acknowledges via FA to MN
- o these actions have to be secured by authentication

❑ Advertisement

- o HA advertises the IP address of the MN (as for fixed systems), i.e. standard routing information
- o routers adjust their entries, these are stable for a longer time (HA responsible for a MN over a longer period of time)
- o packets to the MN are sent to the HA,
- o independent of changes in COA/FA

Agent advertisement

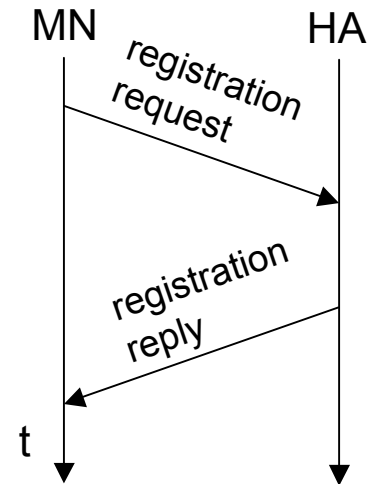
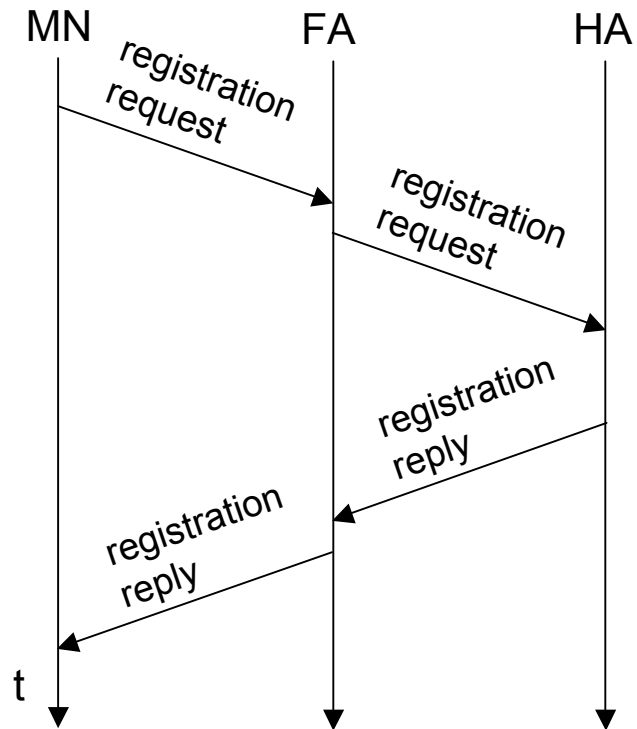
0	7	8	15	16	23	24	31				
type		code		checksum							
#addresses		addr. size		lifetime							
router address 1											
preference level 1											
router address 2											
preference level 2											
...											
...											
type		length		sequence number							
registration lifetime				R	B	H	F	M	G	V	reserved
COA 1											
COA 2											
...											

R: registration required
 B: busy
 H: home agent
 F: foreign agent
 M: minimal encapsulation
 G: generic encapsulation
 V: header compression

ICMP-Type = 0; Code = 0/16; Extension Type = 16

TTL = 1 Dest-Adr = 224.0.0.1 (multicast on link) or 255.255.255.255 (broadcast)

Registration



Goal: inform the home agent of current location of MN (COA-FA or co-located COA)

Registration expires automatically (lifetime)

Uses UDP port 434

Mobile IP registration request

0	7	8	15	16	23	24	31			
type		S	B	D	M	G	V	rsv	lifetime	
home address										
home agent										
COA										
identification										
extensions . . .										

UDP packet on port 343

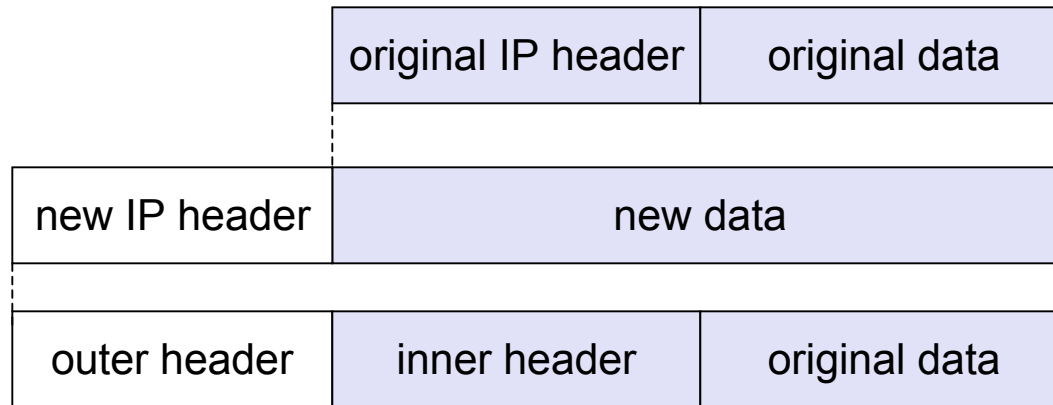
Type = 1 for registration request

S: retain prior mobility bindings

B: forward broadcast packets

D: co-located address=> MN decapsulates packets

Encapsulation



Encapsulation I

- ❑ Encapsulation of one packet into another as payload
 - o e.g. IPv6 in IPv4 (6Bone), Multicast in Unicast (Mbone)
 - o here: e.g. IP-in-IP-encapsulation, minimal encapsulation or GRE (Generic Record Encapsulation)
- ❑ IP-in-IP-encapsulation (mandatory in RFC 2003)
 - o tunnel between HA and COA

ver.	IHL	TOS	length	
IP identification		flags	fragment offset	
TTL		<i>IP-in-IP</i>	IP checksum	
IP address of HA				
Care-of address COA				
ver.	IHL	TOS	length	
IP identification		flags	fragment offset	
TTL		lay. 4 prot.	IP checksum	
IP address of CN				
IP address of MN				
TCP/UDP/ ... payload				

Encapsulation II

- ❑ Minimal encapsulation (optional) [RFC2004]
 - o avoids repetition of identical fields
 - o e.g. TTL, IHL, version, TOS
 - o only applicable for unfragmented packets, no space left for fragment identification

ver.	IHL	TOS	length	
IP identification			flags	fragment offset
TTL	<i>min. encap.</i>		IP checksum	
IP address of HA				
care-of address COA				
lay. 4 protoc.	S	reserved	IP checksum	
IP address of MN				
original sender IP address (if S=1)				
TCP/UDP/ ... payload				

Optimization of packet forwarding

❑ Triangular Routing

- o sender sends all packets via HA to MN
- o higher latency and network load

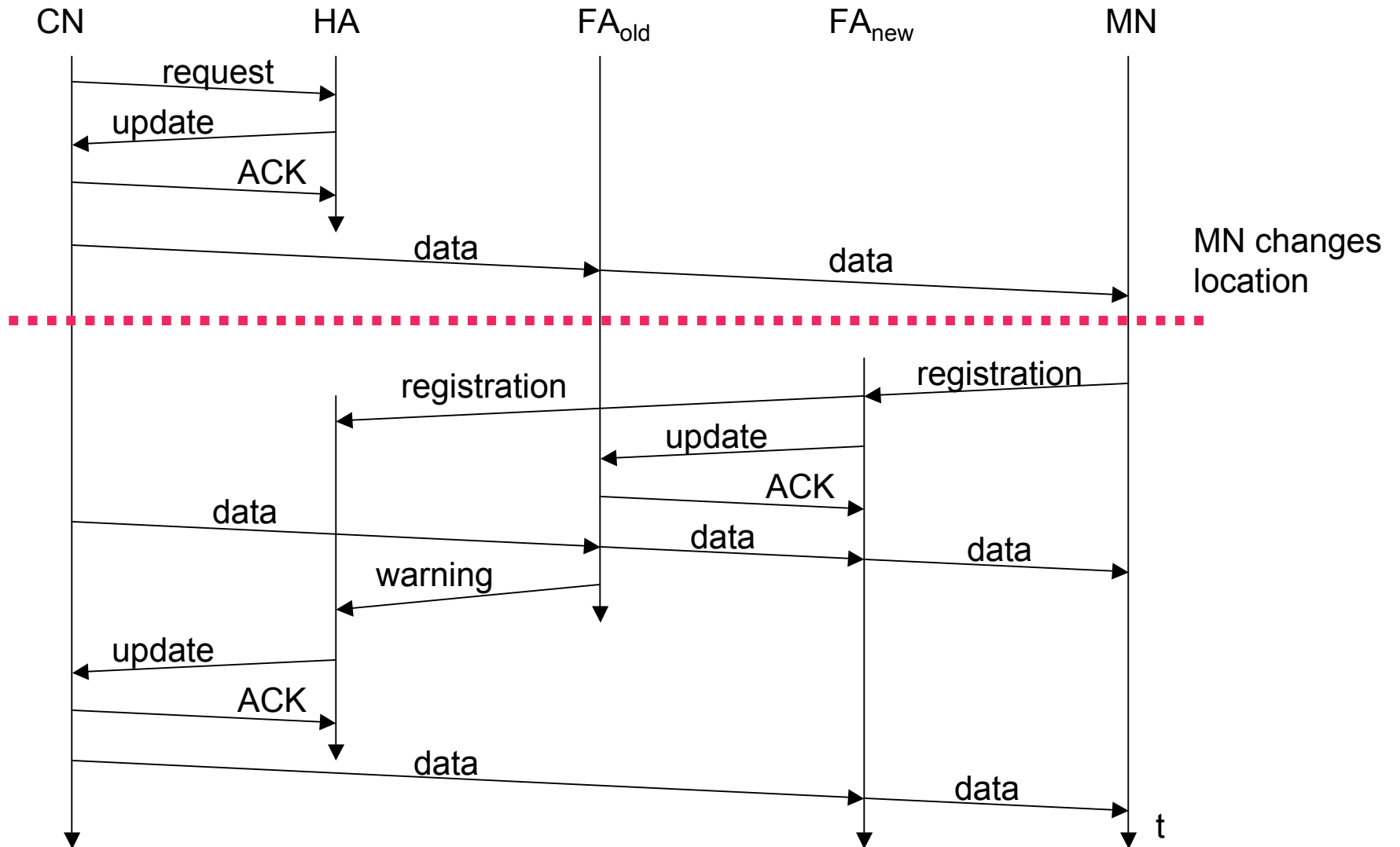
❑ “Solutions”

- o sender learns the current location of MN
- o direct tunneling to this location
- o HA informs a sender about the location of MN
- o big security problems!

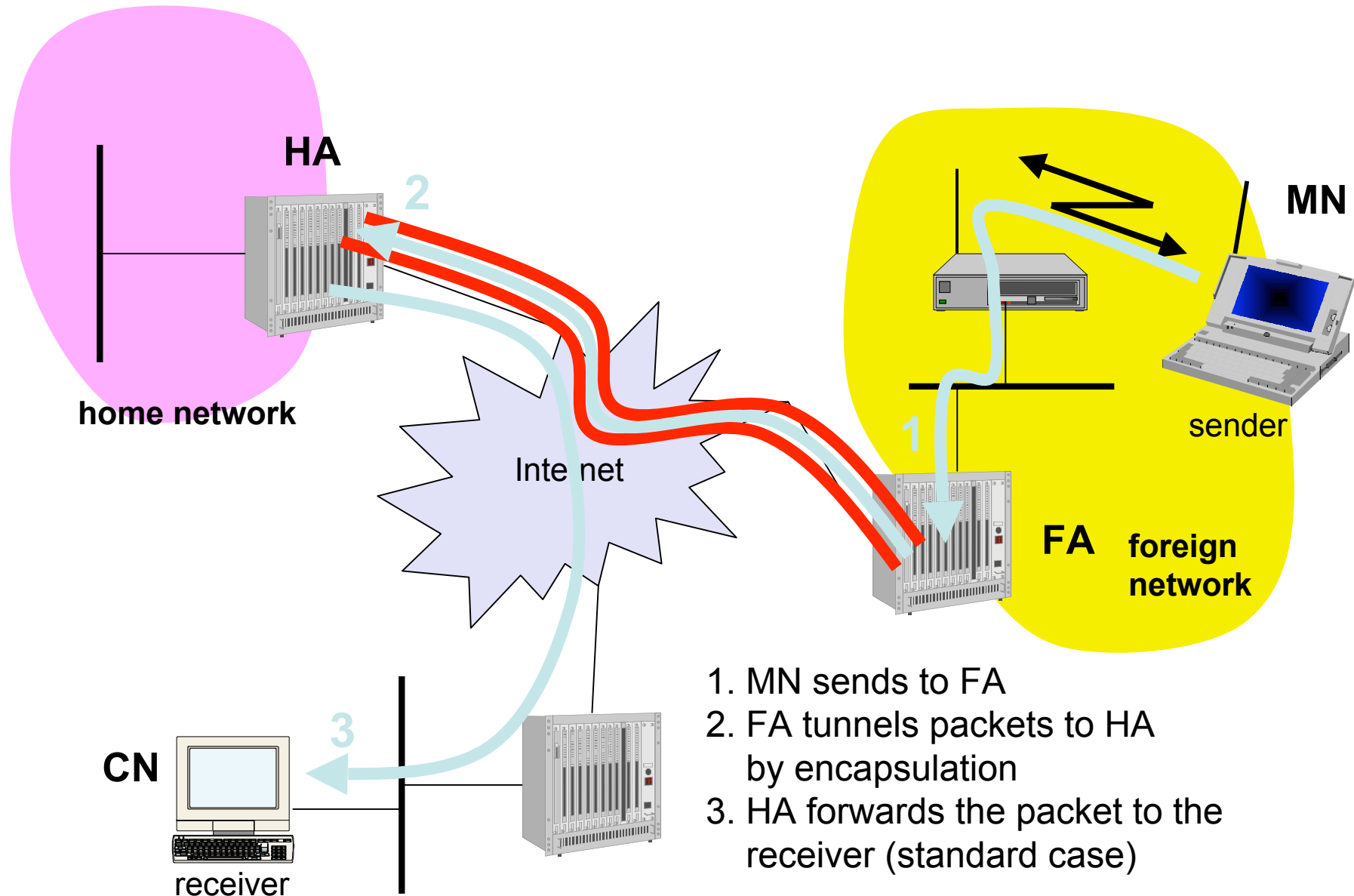
❑ Change of FA

- o packets on-the-fly during the change can be lost
- o new FA informs old FA to avoid packet loss, old FA now forwards remaining packets to new FA
- o this information also enables the old FA to release resources for the MN

Change of foreign agent



Reverse tunneling (RFC 2344)



Mobile IP with reverse tunneling

- ❑ Routers accept often only “topological correct” addresses (firewall)
 - o a packet from the MN encapsulated by the FA is now topological correct
 - o furthermore multicast and TTL problems solved (TTL in the home network correct, but MN is too far away from the receiver)
- ❑ Reverse tunneling does not solve
 - o problems with *firewalls*, the reverse tunnel can be abused to circumvent security mechanisms (tunnel hijacking)
 - o optimization of data paths, i.e. packets will be forwarded through the tunnel via the HA to a sender (double triangular routing)
- ❑ The new standard is backwards compatible
 - o the extensions can be implemented easily and cooperate with current implementations without these extensions

Mobile IP and IPv6

- ❑ security is integrated and not an add-on, authentication of registration is included
- ❑ COA can be assigned via auto-configuration (DHCPv6 is one candidate), every node has address autoconfiguration
- ❑ no need for a separate FA, **all** routers perform router advertisement which can be used instead of the special agent advertisement
- ❑ MN can signal a sender directly the COA, sending via HA not needed in this case (automatic path optimization)
- ❑ “soft” hand-over, i.e. without packet loss, between two subnets is supported
 - o MN sends the new COA to its old router
 - o the old router encapsulates all incoming packets for the MN and forwards them to the new COA
 - o authentication is always granted

Problems with Mobile IP

❑ Security

- o authentication with FA problematic, for the FA typically belongs to another organization
- o no protocol for key management and key distribution has been standardized in the Internet
- o patent and export restrictions

❑ Firewalls

- o typically mobile IP cannot be used together with firewalls, special set-ups are needed (such as reverse tunneling)

❑ QoS

- o many new reservations in case of RSVP
- o tunneling makes it hard to give a flow of packets a special treatment needed for the QoS

❑ Security, firewalls, QoS etc. are topics of current research and discussions!